GenMol / UDLM engineering study through V12

30 sampling configurations; 60 scheduled runs; 4704 independently rescored accepted requests out of 4704 planned. Status: complete. These are sampling configurations, not independent trained models.

No superiority established. All engineering settings use two seeds: V5/V6 request 64 per seed, V9/V10/V12 request 100 per seed. The local MDLM reference uses 3 x 1,000; the paper reports 3 runs of 1,000. These different training and sampling budgets are context, not a matched superiority test. Final UDLM seeds 0/1/2 remain reserved.

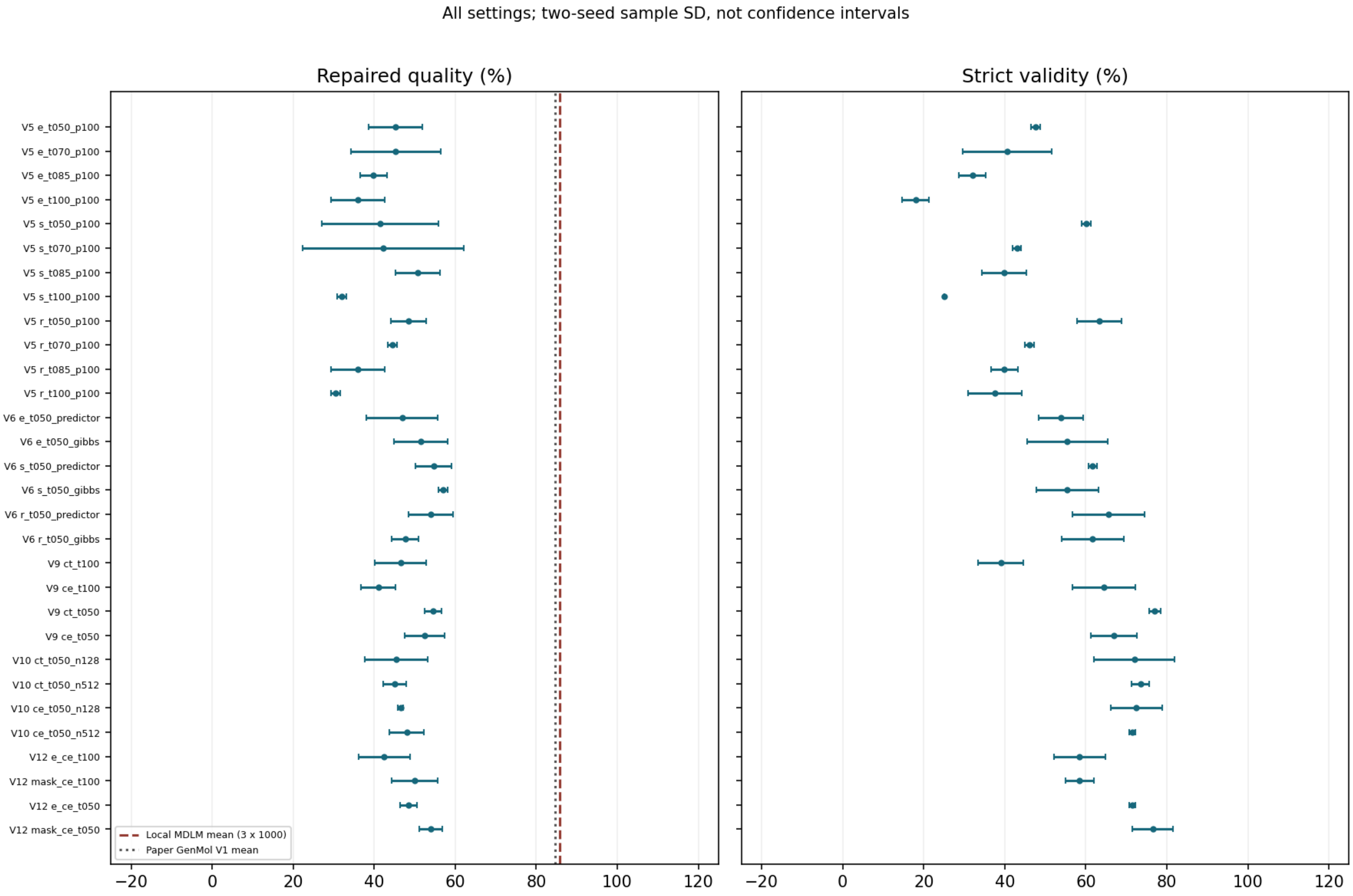
Study	Configs	Scheduled runs	Accepted / planned requests	Run status counts
V5	12	24	1536 / 1536	{'completed': 24}
V6	6	12	768 / 768	{'completed': 12}
V9	4	8	800 / 800	{'completed': 8}
V10	4	8	800 / 800	{'completed': 8}
V12	4	8	800 / 800	{'completed': 8}

The separate failed V4 diagnostic requested 32 samples and remains outside these accepted totals. Original V8 CT controller failure and V11 prelaunch failure remain failures; their follow-ups and the separate CT checkpoint audit retain distinct evidence.

Quality counts within-seed unique valid molecules with QED >= 0.6 and SA <= 4 divided by all requested samples. QED measures drug-likeness (higher is favored); SA estimates synthetic accessibility (lower is favored). Validity uses requests; uniqueness uses valid samples; diversity is the released PyTDC metric on unique valid molecules. Strict SAFE decoding disables repair; released-compatible decoding repairs and selects the largest component. Both start after tokenizer special-token removal.

Every mean and +/- value below is an equal-seed mean and sample SD, not a confidence interval. Means are withheld unless both declared seeds completed and defined the metric. Settings and temperatures were selected adaptively across studies; changed-seed comparisons are not paired treatment effects. Preserved appendices retain their original historical statements and dates.

All configurations: quality and strict validity



V5: every sampling configuration

Configuration / T / kernel	Accepted / planned; seeds	Branch	Validity %	Uniqueness %	Quality %	Diversity
e_t050_p100 / 0.5 / 128P	128/128; [1200, 1201]	Repaired	97.656 +/- 1.105	100.000 +/- 0.000	45.312 +/- 6.629	0.892 +/- 0.006
e_t050_p100 / 0.5 / 128P	128/128; [1200, 1201]	Strict	47.656 +/- 1.105	100.000 +/- 0.000	22.656 +/- 5.524	0.878 +/- 0.002
e_t070_p100 / 0.7 / 128P	128/128; [1200, 1201]	Repaired	98.438 +/- 0.000	99.206 +/- 1.122	45.312 +/- 11.049	0.899 +/- 0.003
e_t070_p100 / 0.7 / 128P	128/128; [1200, 1201]	Strict	40.625 +/- 11.049	100.000 +/- 0.000	21.094 +/- 9.944	0.872 +/- 0.015
e_t085_p100 / 0.85 / 128P	128/128; [1200, 1201]	Repaired	94.531 +/- 1.105	96.680 +/- 2.376	39.844 +/- 3.315	0.904 +/- 0.005
e_t085_p100 / 0.85 / 128P	128/128; [1200, 1201]	Strict	32.031 +/- 3.315	100.000 +/- 0.000	13.281 +/- 3.315	0.883 +/- 0.010
e_t100_p100 / 1.0 / 128P	128/128; [1200, 1201]	Repaired	93.750 +/- 4.419	99.138 +/- 1.219	35.938 +/- 6.629	0.920 +/- 0.012
e_t100_p100 / 1.0 / 128P	128/128; [1200, 1201]	Strict	17.969 +/- 3.315	100.000 +/- 0.000	9.375 +/- 4.419	0.884 +/- 0.007
s_t050_p100 / 0.5 / 128P	128/128; [1200, 1201]	Repaired	97.656 +/- 1.105	99.194 +/- 1.140	41.406 +/- 14.363	0.897 +/- 0.004
s_t050_p100 / 0.5 / 128P	128/128; [1200, 1201]	Strict	60.156 +/- 1.105	100.000 +/- 0.000	23.438 +/- 8.839	0.878 +/- 0.004
s_t070_p100 / 0.7 / 128P	128/128; [1200, 1201]	Repaired	97.656 +/- 1.105	100.000 +/- 0.000	42.188 +/- 19.887	0.898 +/- 0.006
s_t070_p100 / 0.7 / 128P	128/128; [1200, 1201]	Strict	42.969 +/- 1.105	100.000 +/- 0.000	21.094 +/- 1.105	0.864 +/- 0.004
s_t085_p100 / 0.85 / 128P	128/128; [1200, 1201]	Repaired	96.094 +/- 1.105	100.000 +/- 0.000	50.781 +/- 5.524	0.904 +/- 0.004
s_t085_p100 / 0.85 / 128P	128/128; [1200, 1201]	Strict	39.844 +/- 5.524	100.000 +/- 0.000	24.219 +/- 3.315	0.871 +/- 0.004
s_t100_p100 / 1.0 / 128P	128/128; [1200, 1201]	Repaired	92.969 +/- 7.734	98.413 +/- 2.245	32.031 +/- 1.105	0.924 +/- 0.005
s_t100_p100 / 1.0 / 128P	128/128; [1200, 1201]	Strict	25.000 +/- 0.000	100.000 +/- 0.000	11.719 +/- 1.105	0.886 +/- 0.002
r_t050_p100 / 0.5 / 128P	128/128; [1200, 1201]	Repaired	99.219 +/- 1.105	100.000 +/- 0.000	48.438 +/- 4.419	0.881 +/- 0.001
r_t050_p100 / 0.5 / 128P	128/128; [1200, 1201]	Strict	63.281 +/- 5.524	100.000 +/- 0.000	29.688 +/- 4.419	0.869 +/- 0.001
r_t070_p100 / 0.7 / 128P	128/128; [1200, 1201]	Repaired	97.656 +/- 3.315	100.000 +/- 0.000	44.531 +/- 1.105	0.894 +/- 0.002
r_t070_p100 / 0.7 / 128P	128/128; [1200, 1201]	Strict	46.094 +/- 1.105	100.000 +/- 0.000	19.531 +/- 5.524	0.875 +/- 0.002
r_t085_p100 / 0.85 / 128P	128/128; [1200, 1201]	Repaired	96.875 +/- 2.210	99.206 +/- 1.122	35.938 +/- 6.629	0.911 +/- 0.002
r_t085_p100 / 0.85 / 128P	128/128; [1200, 1201]	Strict	39.844 +/- 3.315	100.000 +/- 0.000	21.094 +/- 5.524	0.877 +/- 0.010
r_t100_p100 / 1.0 / 128P	128/128; [1200, 1201]	Repaired	93.750 +/- 0.000	99.167 +/- 1.179	30.469 +/- 1.105	0.911 +/- 0.005
r_t100_p100 / 1.0 / 128P	128/128; [1200, 1201]	Strict	37.500 +/- 6.629	100.000 +/- 0.000	14.062 +/- 2.210	0.881 +/- 0.003

Full inference dictionaries, checkpoint/source identities and per-seed outcomes remain in the appended study reports and overview.json. Missing means are not zero.

V6: every sampling configuration

Configuration / T / kernel	Accepted / planned; seeds	Branch	Validity %	Uniqueness %	Quality %	Diversity
e_t050_predictor / 0.5 / 128P	128/128; [1300, 1301]	Repaired	96.875 +/- 0.000	99.194 +/- 1.140	46.875 +/- 8.839	0.896 +/- 0.010
e_t050_predictor / 0.5 / 128P	128/128; [1300, 1301]	Strict	53.906 +/- 5.524	100.000 +/- 0.000	31.250 +/- 2.210	0.887 +/- 0.006
e_t050_gibbs / 0.5 / 64P+64G	128/128; [1300, 1301]	Repaired	98.438 +/- 2.210	98.412 +/- 0.036	51.562 +/- 6.629	0.889 +/- 0.003
e_t050_gibbs / 0.5 / 64P+64G	128/128; [1300, 1301]	Strict	55.469 +/- 9.944	100.000 +/- 0.000	29.688 +/- 0.000	0.874 +/- 0.003
s_t050_predictor / 0.5 / 128P	128/128; [1300, 1301]	Repaired	98.438 +/- 2.210	99.194 +/- 1.140	54.688 +/- 4.419	0.882 +/- 0.006
s_t050_predictor / 0.5 / 128P	128/128; [1300, 1301]	Strict	61.719 +/- 1.105	100.000 +/- 0.000	33.594 +/- 1.105	0.866 +/- 0.007
s_t050_gibbs / 0.5 / 64P+64G	128/128; [1300, 1301]	Repaired	98.438 +/- 2.210	97.606 +/- 1.176	57.031 +/- 1.105	0.892 +/- 0.005
s_t050_gibbs / 0.5 / 64P+64G	128/128; [1300, 1301]	Strict	55.469 +/- 7.734	100.000 +/- 0.000	36.719 +/- 1.105	0.878 +/- 0.004
r_t050_predictor / 0.5 / 128P	128/128; [1300, 1301]	Repaired	100.000 +/- 0.000	98.438 +/- 2.210	53.906 +/- 5.524	0.885 +/- 0.007
r_t050_predictor / 0.5 / 128P	128/128; [1300, 1301]	Strict	65.625 +/- 8.839	100.000 +/- 0.000	36.719 +/- 3.315	0.874 +/- 0.001
r_t050_gibbs / 0.5 / 64P+64G	128/128; [1300, 1301]	Repaired	97.656 +/- 1.105	100.000 +/- 0.000	47.656 +/- 3.315	0.893 +/- 0.012
r_t050_gibbs / 0.5 / 64P+64G	128/128; [1300, 1301]	Strict	61.719 +/- 7.734	100.000 +/- 0.000	32.031 +/- 5.524	0.877 +/- 0.008

Full inference dictionaries, checkpoint/source identities and per-seed outcomes remain in the appended study reports and overview.json. Missing means are not zero.

V9: every sampling configuration

Configuration / T / kernel	Accepted / planned; seeds	Branch	Validity %	Uniqueness %	Quality %	Diversity
ct_t100 / 1.0 / 128P	200/200; [1600, 1601]	Repaired	97.500 +/- 2.121	99.479 +/- 0.737	46.500 +/- 6.364	0.909 +/- 0.000
ct_t100 / 1.0 / 128P	200/200; [1600, 1601]	Strict	39.000 +/- 5.657	100.000 +/- 0.000	21.000 +/- 1.414	0.886 +/- 0.007
ce_t100 / 1.0 / 128P	200/200; [1600, 1601]	Repaired	98.000 +/- 1.414	99.485 +/- 0.729	41.000 +/- 4.243	0.903 +/- 0.004
ce_t100 / 1.0 / 128P	200/200; [1600, 1601]	Strict	64.500 +/- 7.778	100.000 +/- 0.000	26.000 +/- 7.071	0.887 +/- 0.005
ct_t050 / 0.5 / 128P	200/200; [1600, 1601]	Repaired	98.500 +/- 0.707	99.495 +/- 0.714	54.500 +/- 2.121	0.884 +/- 0.003
ct_t050 / 0.5 / 128P	200/200; [1600, 1601]	Strict	77.000 +/- 1.414	100.000 +/- 0.000	44.000 +/- 1.414	0.868 +/- 0.008
ce_t050 / 0.5 / 128P	200/200; [1600, 1601]	Repaired	99.000 +/- 0.000	99.495 +/- 0.714	52.500 +/- 4.950	0.888 +/- 0.006
ce_t050 / 0.5 / 128P	200/200; [1600, 1601]	Strict	67.000 +/- 5.657	100.000 +/- 0.000	39.500 +/- 3.536	0.870 +/- 0.001

Full inference dictionaries, checkpoint/source identities and per-seed outcomes remain in the appended study reports and overview.json. Missing means are not zero.

V10: every sampling configuration

Configuration / T / kernel	Accepted / planned; seeds	Branch	Validity %	Uniqueness %	Quality %	Diversity
ct_t050_n128 / 0.5 / 128P	200/200; [1700, 1701]	Repaired	100.000 +/- 0.000	100.000 +/- 0.000	45.500 +/- 7.778	0.892 +/- 0.006
ct_t050_n128 / 0.5 / 128P	200/200; [1700, 1701]	Strict	72.000 +/- 9.899	100.000 +/- 0.000	35.500 +/- 12.021	0.875 +/- 0.002
ct_t050_n512 / 0.5 / 512P	200/200; [1700, 1701]	Repaired	99.500 +/- 0.707	100.000 +/- 0.000	45.000 +/- 2.828	0.891 +/- 0.001
ct_t050_n512 / 0.5 / 512P	200/200; [1700, 1701]	Strict	73.500 +/- 2.121	100.000 +/- 0.000	32.500 +/- 3.536	0.876 +/- 0.005
ce_t050_n128 / 0.5 / 128P	200/200; [1700, 1701]	Repaired	99.000 +/- 1.414	99.490 +/- 0.722	46.500 +/- 0.707	0.893 +/- 0.003
ce_t050_n128 / 0.5 / 128P	200/200; [1700, 1701]	Strict	72.500 +/- 6.364	100.000 +/- 0.000	36.000 +/- 0.000	0.878 +/- 0.001
ce_t050_n512 / 0.5 / 512P	200/200; [1700, 1701]	Repaired	98.500 +/- 0.707	100.000 +/- 0.000	48.000 +/- 4.243	0.893 +/- 0.001
ce_t050_n512 / 0.5 / 512P	200/200; [1700, 1701]	Strict	71.500 +/- 0.707	100.000 +/- 0.000	38.500 +/- 3.536	0.878 +/- 0.005

Full inference dictionaries, checkpoint/source identities and per-seed outcomes remain in the appended study reports and overview.json. Missing means are not zero.

V12: every sampling configuration

Configuration / T / kernel	Accepted / planned; seeds	Branch	Validity %	Uniqueness %	Quality %	Diversity
e_ce_t100 / 1.0 / 128P	200/200; [2000, 2001]	Repaired	99.000 +/- 1.414	99.500 +/- 0.707	42.500 +/- 6.364	0.901 +/- 0.003
e_ce_t100 / 1.0 / 128P	200/200; [2000, 2001]	Strict	58.500 +/- 6.364	100.000 +/- 0.000	27.500 +/- 2.121	0.885 +/- 0.006
mask_ce_t100 / 1.0 / 128P	200/200; [2000, 2001]	Repaired	99.500 +/- 0.707	99.500 +/- 0.707	50.000 +/- 5.657	0.905 +/- 0.000
mask_ce_t100 / 1.0 / 128P	200/200; [2000, 2001]	Strict	58.500 +/- 3.536	100.000 +/- 0.000	34.000 +/- 5.657	0.882 +/- 0.001
e_ce_t050 / 0.5 / 128P	200/200; [2000, 2001]	Repaired	98.000 +/- 1.414	100.000 +/- 0.000	48.500 +/- 2.121	0.894 +/- 0.007
e_ce_t050 / 0.5 / 128P	200/200; [2000, 2001]	Strict	71.500 +/- 0.707	100.000 +/- 0.000	37.500 +/- 3.536	0.883 +/- 0.008
mask_ce_t050 / 0.5 / 128P	200/200; [2000, 2001]	Repaired	99.500 +/- 0.707	99.495 +/- 0.714	54.000 +/- 2.828	0.886 +/- 0.004
mask_ce_t050 / 0.5 / 128P	200/200; [2000, 2001]	Strict	76.500 +/- 4.950	100.000 +/- 0.000	42.000 +/- 4.243	0.876 +/- 0.000

Full inference dictionaries, checkpoint/source identities and per-seed outcomes remain in the appended study reports and overview.json. Missing means are not zero.

Training outcomes remain separate

Training event	Outcome	Updates / configured exposures	Evidence / interpretation
Historical E	Audited frozen training checkpoint	1000 / 16000	Batch16; seed17; checkpoint dce870e8d634
Historical R	Audited frozen training checkpoint	1000 / 16000	Batch16; seed17; checkpoint d0310d2e2402
Historical S	Audited frozen training checkpoint	1000 / 16000	Batch16; seed17; checkpoint 100f467b9476
V7 throughput pilot	completed	20 / 2560	Source e78267b09810; checkpoint 07ad66498cca; final finite tensors 1155
V8 CT controller	failed	not accepted by controller / None	Source c8434dda105c; separate post-exit audit accepted step1000, checkpoint 48986899c401; failed receipt unchanged
V8b empirical CE	completed	1000 / 128000	Source b84f21ba1f30; checkpoint b7d674ed5bdd; final finite tensors 1156
V11 MASK-rich CE	failed	not accepted by controller / None	Source 46ec3b0bc0c5; no child or checkpoint
V11b MASK-rich CE	completed	1000 / 128000	Source be18a3244a71; checkpoint 62299de99d8c; final finite tensors 1157
Original V8 campaign	failed	CT then campaign stopped	Original CE arm never launched; V8b is a separate fresh follow-up.

All continuations start from original MDLM 50k EMA with fresh optimizer/EMA. Historical R/S/E add 1,000 updates at batch 16 (16,000 exposures). V8 CT, V8b CE and V11b CE add 1,000 at batch 128 (128,000 exposures), eight times as many exposures. Exposures are not necessarily distinct molecules. Final checkpoint finiteness does not prove every update was finite. Original MDLM 50k training has a different budget.

V12: signed MASK-rich minus empirical CE contrasts

Contrast / T	Branch / metric	Seed2000 delta	Seed2001 delta	Mean +/- sample SD
primary / 1.0	released_comparable / validity	+0.000	+1.000	+0.500 +/- 0.707
primary / 1.0	released_comparable / uniqueness	+0.000	+0.000	+0.000 +/- 0.000
primary / 1.0	released_comparable / quality	+16.000	-1.000	+7.500 +/- 12.021
primary / 1.0	released_comparable / diversity	+0.006	+0.001	+0.004 +/- 0.004
primary / 1.0	strict / validity	-2.000	+2.000	+0.000 +/- 2.828
primary / 1.0	strict / uniqueness	+0.000	+0.000	+0.000 +/- 0.000
primary / 1.0	strict / quality	+12.000	+1.000	+6.500 +/- 7.778
primary / 1.0	strict / diversity	+0.001	-0.007	-0.003 +/- 0.005
secondary / 0.5	released_comparable / validity	+1.000	+2.000	+1.500 +/- 0.707
secondary / 0.5	released_comparable / uniqueness	+0.000	-1.010	-0.505 +/- 0.714
secondary / 0.5	released_comparable / quality	+6.000	+5.000	+5.500 +/- 0.707
secondary / 0.5	released_comparable / diversity	-0.006	-0.010	-0.008 +/- 0.003
secondary / 0.5	strict / validity	+8.000	+2.000	+5.000 +/- 4.243
secondary / 0.5	strict / uniqueness	+0.000	+0.000	+0.000 +/- 0.000
secondary / 0.5	strict / quality	+5.000	+4.000	+4.500 +/- 0.707
secondary / 0.5	strict / diversity	-0.001	-0.012	-0.007 +/- 0.008

Percentage-point differences for validity, uniqueness and quality; raw units for diversity. Pairing is by declared seed, not molecular trajectory. A failed seed withholds the paired mean/SD. Temperature 1 is primary; 0.5 is secondary and acts after CE-to-LOO conversion.

V12: final editable MASK occurrences

Configuration	Seed	Final MASK / editable positions	Rows with MASK / requests	All generated control counts
e_ce_t100	2000	0 / 4789	0 / 100	{'unk': 0, 'bos': 0, 'eos': 0, 'pad': 0, 'mask': 0}
e_ce_t100	2001	0 / 4837	0 / 100	{'unk': 0, 'bos': 0, 'eos': 0, 'pad': 0, 'mask': 0}
mask_ce_t100	2000	0 / 4789	0 / 100	{'unk': 0, 'bos': 0, 'eos': 0, 'pad': 0, 'mask': 0}
mask_ce_t100	2001	0 / 4837	0 / 100	{'unk': 0, 'bos': 0, 'eos': 0, 'pad': 0, 'mask': 0}
e_ce_t050	2000	0 / 4789	0 / 100	{'unk': 0, 'bos': 0, 'eos': 0, 'pad': 0, 'mask': 0}
e_ce_t050	2001	0 / 4837	0 / 100	{'unk': 0, 'bos': 0, 'eos': 0, 'pad': 0, 'mask': 0}
mask_ce_t050	2000	0 / 4789	0 / 100	{'unk': 0, 'bos': 0, 'eos': 0, 'pad': 0, 'mask': 0}
mask_ce_t050	2001	0 / 4837	0 / 100	{'unk': 0, 'bos': 0, 'eos': 0, 'pad': 0, 'mask': 0}

Counts come from hash-validated final uint16 IDs and the saved editable bitmask; padding/framing are excluded. CSV text skips special tokens, so high strict chemical validity does not certify zero generated controls. These are final occurrences, not survival trajectories. The ideal clean-content forward endpoint at lambda 0.9 has approximately 9 ppm MASK probability at positive epsilon 1e-5; learned predictions, finite-step factorization, temperature and independent-prior initialization prevent using that as an empirical guarantee.

The manifest hashes every retained input, current imported helper source, archived generator and output. Large checkpoint bytes are not reloaded here: their digests and prior/CE/EMA acceptance are inherited from the bound CPU validation/training and independently rescored generation receipts. Original PDFs follow unchanged, beginning with the 43-page V9 overview.